

Racetrack Assignment: Neural Network (Optional Extension)

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1 Overview

Note: This portion of the Racetrack project is optional and ungraded. You may complete it if you wish to explore neural network control using Unity ML-Agents.

The goal is to train a neural network to drive the racecar autonomously using reinforcement learning. You will use Unity's **ml-agents** package, which provides the necessary tools to train and evaluate agents in simulated environments.

2 Setup

- Start with `NNVehicle.cs` under `Scripts/GameAIStudentWork/`.
- Open the `RacetrackNN` scene in Unity.
- Complete the script by defining the observations and the training policy, then follow the ML-Agents training workflow.

Refer to the official Unity ML-Agents documentation for installation and setup:

- [Unity ML-Agents 4.0 Manual](#)
- [ML-Agents Release 23 Notes](#)

The Unity project already includes the **ML-Agents 4.0.0** package.

3 Training Tips and Resources

It is strongly recommended that you read through the Unity ML-Agents introductory documentation before beginning. The following links may assist you in configuring and training your agent:

Algorithm Background

- [Proximal Policy Optimization \(PPO\)](#) — overview of the reinforcement learning algorithm used by ML-Agents.

Network Architecture

- [Configuring Layers and Nodes in a Neural Network](#) — guidelines for selecting network size and depth.

Reward Design

- [ML-Agents: Environment and Reward Design](#)

Performance and Optimization

- [Using an Environment Executable](#) — speeds up training by running Unity headless or in low-graphics mode.
- Reduce graphics load by setting Unity to windowed mode with low resolution.
- You may need to increase the communication buffer size for concurrent environments: [Unity Forum: Concurrent Environment Discussion](#).
- [Training Using Concurrent Unity Instances](#) — run multiple simulations in parallel to accelerate learning.

4 Summary

This optional task allows you to experiment with neural network control and compare it against fuzzy logic control. Successful training may require tuning the observation set, rewards, and training hyperparameters. While not graded, this experience can provide valuable insight into reinforcement learning methods and their tradeoffs in dynamic control tasks such as the Racetrack environment.